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Optimization strategies to mitigate reactant starvation in a dead-ended hydrogen–oxygen proton exchange membrane fuel cell during cyclic loading

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ABSTRACT

Investigating the dynamic characteristics in hydrogen–oxygen proton exchange membrane fuel cells (PEMFCs) can provide valuable insights into the coupling degradation mechanism present within the fuel cell, which can help avoid reactant starvation and voltage decline. In this study, a recirculation subsystem is integrated into the cathode of the PEMFC to alleviate gas starvation due to water flooding. The impact of pump speed on dynamic loading is also examined. Furthermore, obstacle devices are incorporated into the cell flow channel to enhance mass transfer. The effects of flow channel configuration under the exhaust gas recirculation mode on cell performance and dynamic characteristics are investigated as well. Our findings indicate that blockage of the gas transport channel due to accumulated liquid water is the primary issue during the loading process of PEMFCs. The cathode recirculation pumps set at 600 rpm improves cell performance by 30.6% compared to the dead-ended mode at 1000 mA·cm^{-2}. However, excessively high pump speeds lead to an increase in voltage undershoot during the loading process. Exhaust gas recirculation is found to significantly alleviate gas starvation and water flooding during the cyclic loading process and the cathode recirculation is the primary mitigation strategy for improving dynamic characteristics.

1. Introduction

Proton exchange membrane fuel cells (PEMFCs) have become promising devices that directly convert the chemical energy in fuel into electrical energy under the action of a catalyst [1–3]. The PEMFCs can be classified into two types: hydrogen-air fuel cells and hydrogen–oxygen fuel cells. In hydrogen-air fuel cells, the cathode side is supplied with air, readily obtained from the surrounding environment. However, to achieve higher flow rates within the channels, the inclusion of an air compressor becomes necessary in hydrogen-air fuel cell systems, resulting in a decrease in overall system efficiency. Moreover, the operation of the air compressor generates noticeable vibration and noise [4]. On the other hand, hydrogen–oxygen fuel cells employ pure oxygen as the cathode reaction gas, and the absence of moving components throughout the system allows for high energy conversion efficiency and low noise levels. Consequently, hydrogen–oxygen fuel cells hold significant potential for military applications, including unmanned ships, underwater unmanned vehicles, and space power sources [5–7].

However, a serious problem that restricts the development of fuel cells is their lifespan degradation during loading process. A large amount of research on the dynamic characteristics of fuel cells is conducted through experiments. Many scholars have designed various loading experiments for fuel cells by changing the operating parameters and analyzing the results of the loading experiments to study the dynamic characteristics of PEMFCs [8]. To investigate the characteristics inside the fuel cell during dynamic loading, some advanced technologies have also been used in the research of fuel cells' dynamic characteristics, such as segmented fuel cell technology [9,10], visualization technology [11], high-resolution neutron imaging technology [12], and local measurement of current density [13], etc. Jia et al. [14] investigated the dynamic distribution characteristics of fuel and water under oxidant starvation by in-situ measuring the local current density, temperature, and voltage of the fuel cell. Zhang et al. [15] measured the local current density and temperature simultaneously in a PEMFC with a single serpentine flow field during dynamic loading. Besides, to investigate the degradation behavior of fuel cells under dynamic cycles, Wang et al.

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[16] conducted an experimental study of common operating conditions from idling to a rated condition in automotive fuel cells and found that dynamic cycles have a significant negative effect on the catalyst layer, which leads to a significant reduction in catalyst layer thickness. The irreversible degradation of membrane electrode assembly (MEA) due to gas starvation under dynamic conditions of the fuel cell was also reported by Chu et al. [17,18].

Many related studies have been reported to alleviate gas starvation during dynamic loading of the PEMFC. Zhong et al. [19] proposed that the structure of the anode flow field will directly affect gas distribution in fuel cells. In addition, two parallel anode flow field optimization schemes were proposed and experimentally studied. Jia et al. [20] proposed an operational strategy to alleviate hydrogen starvation in fuel cells and investigated its effectiveness by measuring the local current density and temperature under different load change scenarios. To alleviate and eliminate hydrogen starvation during dynamic loading, Bai et al. [21] proposed a dual-path hydrogen supply strategy to improve mass transfer in the stack while enhancing the voltage stability and temperature consistency of the single cell. In addition, Wang et al. [22] investigated the coupled control algorithms for PID control and model predictive control in series and parallel, respectively, to mitigate oxidant starvation during dynamic loading of the fuel cell. Moreover, the application of machine learning in the study of the development of optimization strategies during dynamic loading in the PEMFC have received widespread attention [23–25]. Wang et al. [26] developed a long short-term memory deep learning model for predicting the dynamic performance of PEMFCs. They also optimized parameters such as input feature selection, sampling frequency, and sequence duration. Rui et al. [27] proposed a method based on machine learning to design durable MEA that enable fuel cells to exhibit excellent performance during dynamic responses.

The above studies are all investigations of the dynamic characteristic of the PEMFC fed by hydrogen and air. There are fewer reports on the voltage response characteristics of H_2/O_2 PEMFCs under gas starvation conditions. Meng et al. [28,29] investigated the voltage response characteristics of a H_2/O_2 PEMFC under different levels of gas starvation and gave a quantitative analysis of the hydrogen and oxygen starvation tolerance situation by polynomial fitting. Chen et al. [30] then quantitatively investigated the transient voltage overshoot phenomenon caused by current loading in H_2/O_2 fuel cells. However, a clearer explanation of the hydration mechanism within the fuel cell at the moment of dynamic loading and the coupling degradation mechanism has not been mentioned. The H_2/O_2 fuel cells are usually operated in dead-ended anode and cathode (DEAC) mode, and the severe water management problems will lead to increased uncertainty in the dynamic response characteristics [31,32].

In this study, a recirculation subsystem was added to the cathode of the H_2/O_2 PEMFC to alleviate gas starvation caused by water flooding, and the impact of pump speed on dynamic loading was studied. In addition, the obstacle devices were integrated into the cell flow channel to enhance mass transfer, and the impact of flow channel configuration under the exhaust gas recirculation mode on cell performance and dynamic loading characteristics was also investigated. Based on this systematic research, the optimal configuration for alleviating gas starvation during the fuel cell variable load process was determined, followed by the long-term continuous variable load acceleration testing of the fuel cell MEA. Additionally, polarization curves, electrochemical impedance spectroscopy (EIS), and cyclic voltammetry (CV) were tested for the MEA of the fuel cell under different system configurations after the experiment.

2. Experimental

2.1. Experimental system

The MEA used by Wuhan WUT New Energy Co., Ltd. in the fuel cell was customized and composed of a proton exchange membrane, catalyst layer (CL), and gas diffusion layer (GDL). Nafion® XL from DuPont™ was used as the proton exchange membrane. The catalyst (Hispec™ 9100, Johnson Matthey) was directly sprayed on both sides of the proton exchange membrane, with a cathode platinum loading of 0.3 $mg \cdot cm^{-2}$ and an anode platinum loading of 0.1 $mg \cdot cm^{-2}$. The GDL adopts Toray's TGP-060, which was surface-treated with polytetrafluoroethylene (PTFE) for hydrophobicity with a contact angle of 145°. The flow field structure of a single cell adopts a straight channel, which is carved by graphite board and has undergone hydrophobic treatment on the surface. The processing accuracy is controlled within ± 0.01 mm. The low-resistivity metal copper was used as the current collector plate, and its surface was subjected to gold plating treatment to reduce contact resistance. The cell was placed vertically to facilitate drainage by gravity. The water generated by the cathode can be collected in the buffer channel to avoid flooding the main channel. The geometric parameters of the single cell are shown in Table 1 [33].

Table 1
Geometric parameters of assembled PEMFC.

Parameters	Value	Unit
Electrochemical active area	50×10^{-4}	m^2
Thickness of GDLs	2.5×10^{-4}	m
Thickness of membrane	2.5×10^{-5}	m
Thickness of CL	1.0×10^{-5}	m
Depth of flow channels (Straight channel)	1.0×10^{-3}	m
Width of flow channels (Straight channel)	2.0×10^{-3}	m
Width of ribs (Straight channel)	2.0×10^{-3}	m
Depth of flow channels (Obstacle channel)	1.0×10^{-3}	m
Width of flow channels (Obstacle channel)	2.0×10^{-3}	m
Width of ribs (Obstacle channel)	2.0×10^{-3}	m
Radius of a semi-cylindrical obstacle (Obstacle channel)	5.0×10^{-4}	m
Spacing between semi-cylindrical obstacles (Obstacle channel)	3.0×10^{-3}	m

The performance test of the fuel cell is based on the Hephas 850e multi-range fuel cell test system, which is manufactured by Scribner Associates Inc. This test platform has high stability and a precise fuel control system. It can not only perform performance tests on single cells but also perform cyclic voltammetry analysis on the cell. The schematic diagram of the test system is shown in Fig. 1. High-purity oxygen and hydrogen in the gas tank are controlled at constant pressure and introduced into the cathode and anode through pressure controllers respectively without humidification. To regulate the purging of the cathode and anode outlets, electromagnetic valves are installed. The cell loading current is provided by the 850e. The temperature, voltage, and internal resistance of the cell are monitored by the test system. A buffer tank is set up at the outlet of the cell to collect liquid water during the purging process. The flow field structure of both anode and cathode adopts a straight channel as shown in Fig. 1 (b). In the subsequent optimization of mass transfer enhancement under current loading, the straight channel of cathode side was replaced by the obstacle channel.

The exhaust gas recirculation subsystem in the test system is achieved by a peristaltic pump, which is composed of a driver, pump head, and peristaltic hose. The pressure cover and roller compress the hose while the hose between the two rollers appears as a pillow shape. As the rollers rotate, they alternately squeeze and release the elastic delivery hose to pump the fluid. With the rolling of the roller, the elasticity of the hose continuously restores its original shape, forming a negative pressure at the inlet of the pump to promote gas flow and fill the pipeline for transmission. The peristaltic pumps provide higher control precision and are particularly suited for low-flow rate delivery compared to centrifugal pump. In addition, the object of this study is the hydrogen–oxygen fuel cell used in underwater unmanned vehicles, and the low-noise scene is a factor that should not be ignored, whether it is a kW-level fuel cell stack or a pre-explored single cell. In this study, the BT-100CA series peristaltic pump from Chongqing Jieheng Peristaltic Pumps Co., Ltd. was adopted, with a speed range of 0–600 rpm and a precision of ± 0.1 rpm for speed adjustment.

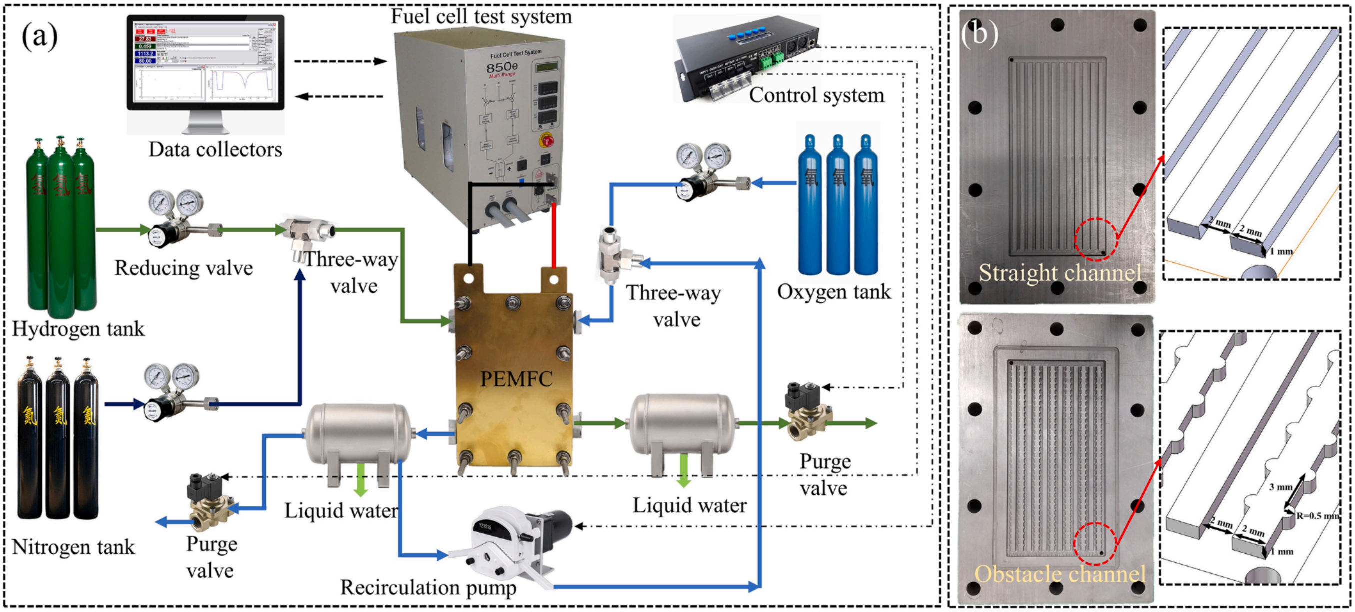

Fig. 1. (a) Schematic diagram of the experimental test system for PEMFC, (b) The flow channel used in the experiment.

2.2. Experimental scheme

The flowchart of the experiment is shown in Fig. 2. Before conducting a dynamic response experiment of fuel cells, the MEA accumulates activation for 4 h with different loading currents. The relative humidity of the inlet fuel remains 100% to ensure that the fuel cell has reached its optimal performance. Before activation, the fuel cell undergoes a meticulous gas tightness test to ensure experimental safety [34]. Moreover, the experiment employs hydrogen and oxygen gases with a purity of 99.999%.

The dynamic response experiment of fuel cells consists of three main steps. Firstly, characteristics of dynamic loading of fuel cells under the DEAC mode were explored, analyzing the influence of the current loading amplitude and cumulative operating time before variable load on the voltage response characteristics. A recirculation subsystem was added to the cathode of the fuel cell to alleviate gas starvation caused by water flooding, and the impact of pump speed on dynamic loading was studied. Following this, to address gas starvation during the variable load process, obstacle devices were integrated into the cell flow channel to enhance mass transfer, and the impact of flow channel configuration under the exhaust gas recirculation mode on cell performance and dynamic loading characteristics was investigated. Based on this systematic research, the optimal configuration for alleviating gas starvation during the fuel cell variable load process was determined, followed by the long-term continuous variable load acceleration testing of the fuel cell MEA. CV testing was conducted on the cell cathode after the experiment, investigating the influence of 500 rounds of cyclic variable load operation of the fuel cell under different system configurations on the electrochemically active surface area of the electrode. During the testing, the anode was supplied with hydrogen gas, while nitrogen gas was supplied to the cathode. Both gases had a flow rate of 200 ml/min and a temperature of 30℃, with a scan rate of 20 mV/s. Additionally, polarization curves and impedance spectra were tested for the MEA of the fuel cell under different system configurations after the experiment.

3. Results and discussions

3.1. Dynamic response characteristics under DEAC mode

The loading amplitude of the fuel cell is an important factor restricting the dynamic response of the cell. Fig. 3 shows the voltage response curves under different loading amplitudes of PEMFC with DEAC mode. In this study, the loading rate was not our main focus and was therefore not considered during experimental testing. The variable loading between different current densities of fuel cells was completed instantaneously. In addition, at t = 0 s, purging actions of the cathode and anode were performed during the experiment to ensure that there was no accumulation of liquid water and impurities gas inside the fuel cell. Before 120 s, the current input was consistently at 200 mA·cm^{-2}, and the voltage was around 0.8 V. At 120 s, the fuel cell was loaded with amplitudes of 400, 800, and 1400 mA·cm^{-2}, respectively. It was observed that during the loading process, the smallest overshoot in fuel cell voltage occurred at the minimum loading amplitude of 400 mA·cm^{-2}, and reached a steady state in the shortest amount of time. As the variable load amplitude increases, the voltage undershoots magnitude will also increase, and it will take longer to return to a steady state. In Fig. 3 (c), under a current loading amplitude of 1400 mA·cm^{-2}, the fuel cell voltage showed significant instability and exhibited a trend of undershoot-recovery-decline. The dynamic response of the fuel cell voltage during variable loads primarily depends on two factors: the diffusion rate of gas in the GDL, and the redistribution of water in the GDL [35]. In the case of large loading amplitude, the response speed of gas transport is much slower than electronic conduction. As a result, the oxygen content in the GDL cannot meet the high current density operation requirements immediately. It takes time for the gas replenishment process to be completed in the GDL, resulting in a voltage drop. The gas-starvation response leads to changes in the moisture content of the GDL, and the process of water redistribution in the MEA requires a certain amount of time [36].

Fig. 4 shows the effect of continuous operation time on voltage response before dynamic loading in DEAC of PEMFC. The duration of continuous operation of the fuel cell before loading determines the difficulty of gas supply and water vapor redistribution during the loading process and also has a significant impact on the dynamic response of the PEMFC. In addition, the dynamic response characteristics of the fuel cell operating in the DEAC mode were explored. We only executed a purging action at t = 0 s. At this time, the solenoid valves at the outlet of both anode and cathode were triggered simultaneously for 2 s. This procedure is to remove the accumulated liquid water and impurity gases in the previous operation to ensure that the subsequent experimental conditions are at the same starting line. It can be seen from the figure that before loading, the voltage of the fuel cell is around 0.675

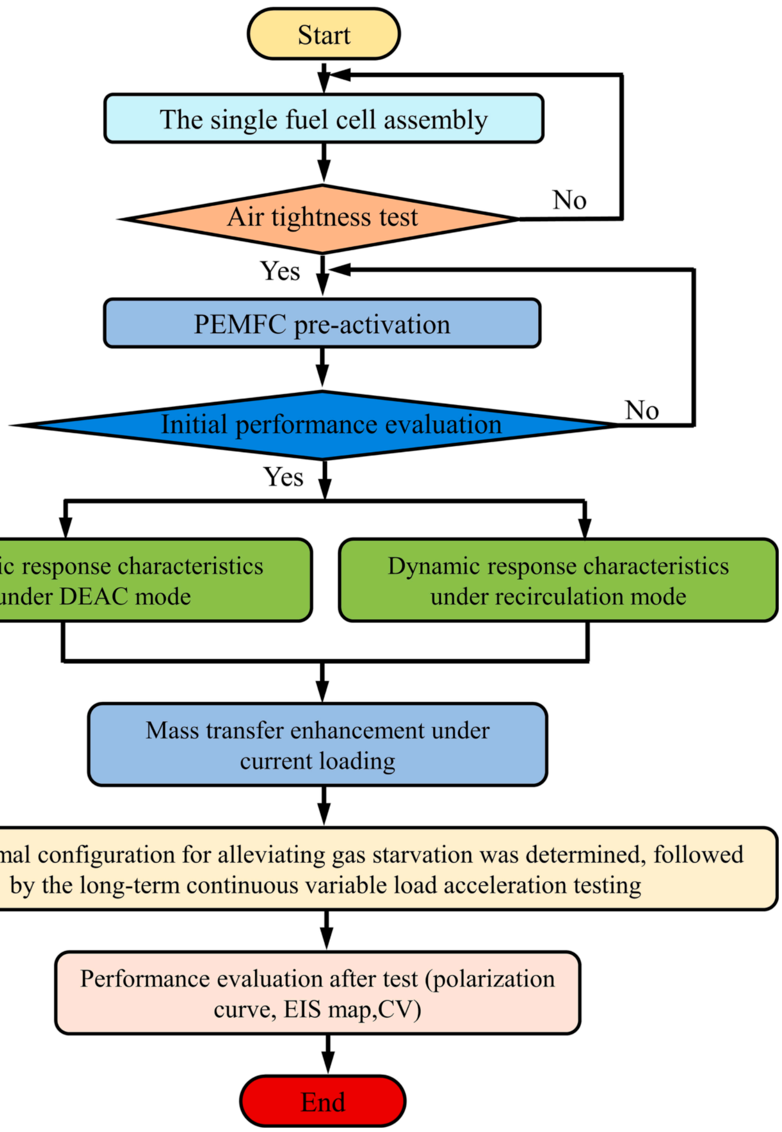

Fig. 2. Flowchart of the experiment.

V, and under the same loading amplitude (from 600 mA·cm^{-2} to 1000 mA·cm^{-2}), the shorter the continuous operation time before loading, the better the performance after loading, and the voltage quickly returns to stability. When the continuous operation time before loading increases to 5 min, liquid water has accumulated inside the fuel cell, causing varying degrees of blockage in the gas diffusion layer. At this point, the large loading amplitude greatly deteriorates the gas supply and water distribution inside the fuel cell, resulting in continuous voltage fluctuations. The fuel cell continues to operate for some time after water flooding, and the voltage continues to drop. From the above study, it can be concluded that the primary issue to be addressed during the loading process of hydrogen–oxygen fuel cells operated in the DEAC mode is the blockage of the GDL gas transport channel caused by accumulated liquid water.

3.2. *Dynamic response characteristics under recirculation mode*

3.2.1. *Effect of recirculation pump speed of cathode side*

In this study, a recirculation pump was installed on the cathode to remove liquid water generated by fuel cells through forced convection. Fig. 5 shows a comparison of the polarization curves of fuel cells at different rotation speeds of the cathode recirculation pump. During the test process, the purging valves at the cathode and anode outlets of the cell were always kept closed. Before testing the polarization curve, the cathode and anode purging valves were triggered to remove liquid water and impurities gas. It can be seen that the cathodic recirculation pump can significantly improve the performance of the fuel cell, especially at high current densities. At a current density of 1000 mA·cm^{-2}, the cathode recirculation pumps of 200 rpm and 600 rpm can improve the cell performance by 19.9% and 30.6%, respectively, compared to the DEAC mode. When the current density was loaded to 1000 mA·cm^{-2}, the performance of the cell operating in the DEAC mode drops sharply due to the increase of liquid water generated by the fuel cell under high current density, which seriously blocks the GDL and prevents the reactant gas from diffusing well to the reaction zone.

Fig. 6 depicts the EIS of the hydrogen–oxygen fuel cell at different pump speeds. To better display the test results, it should be noted that the real part of the EIS initial point does not start from zero. When the cathode recirculation pump speed of the fuel cell increased from 200 to 600 rpm, the ohmic resistance did not change significantly, but the total polarization resistance gradually decreased, thus improving the performance of the cell. At fuel cell current densities of 1000 mA·cm^{-2} and 1600 mA·cm^{-2}, the EIS trend changes were almost consistent, but the higher the current density, the greater the polarization resistance of the

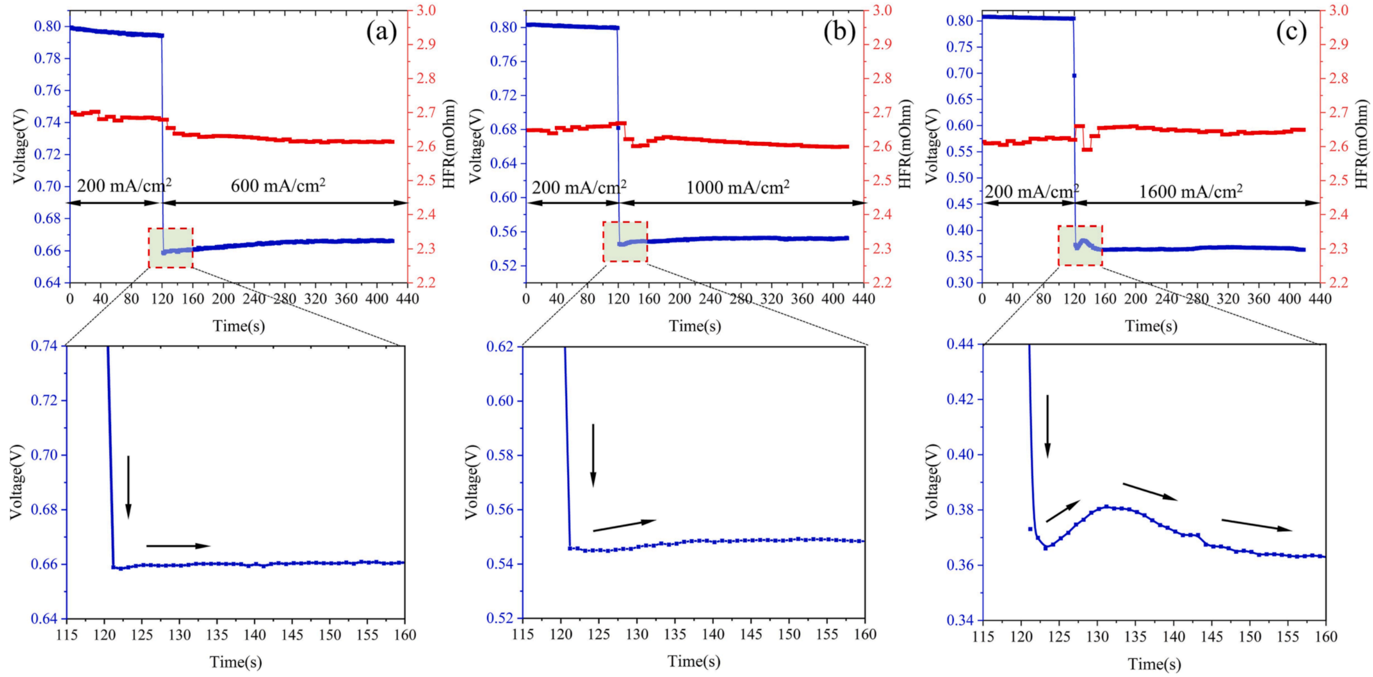

Fig. 3. Voltage response curve under different loading amplitude of PEMFC with DEAC mode. (a) from 200 mA·cm^{-2} to 600 mA·cm^{-2}, (b) from 200 mA·cm^{-2} to 1000 mA·cm^{-2}, (c) from 200 mA·cm^{-2} to 1600 mA·cm^{-2}.

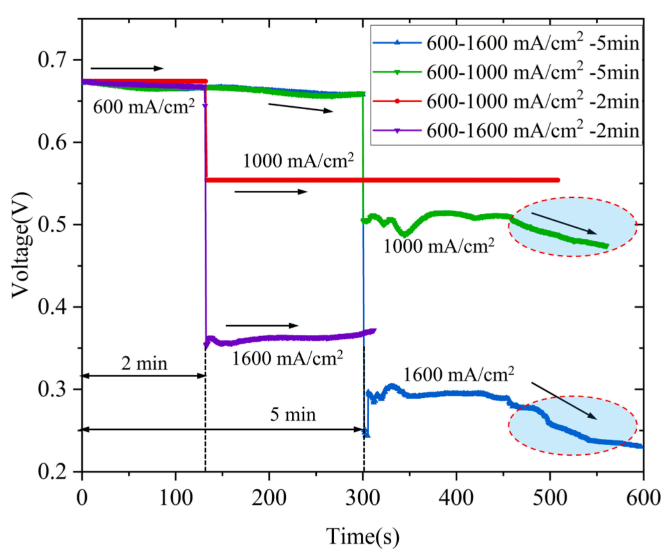

Fig. 4. Effect of continuous operation time on voltage response before dynamic loading in DEAC of PEMFC.

cell.

The voltage response curve from 200 mA·cm^{-2} to 1600 mA·cm^{-2} under different pump speeds of the cathode side was shown in Fig. 7. As shown in Fig. 7 (a), when the fuel cell stably operated at 200 mA·cm^{-2} for 2 min before loading, and since the cathode recirculation pump was always in operation, liquid water hardly accumulated inside the fuel cell. Therefore, the voltage of the cell remained at around 0.84 V. When the current density was suddenly loaded to 1600 mA·cm^{-2}, the cell voltage dropped, but quickly returned to a stable state.

As the speed of the cathode recirculation pump increased, the steady-state voltage value after loading gradually increased. This is because the introduction of the cathode recirculation pump increases the gas flow rate inside the fuel cell, leading to the removal of liquid water generated during the reaction. Therefore, the performance of the fuel cell behaves best at pump speeds of 600 rpm after 230 s. The voltage undershoots amplitudes were 11.8 mV and 5.3 mV at pump speeds of 200 rpm and 400 rpm, respectively, which alleviated gas starvation during loading. However, when the pump speed reached 600 rpm, the voltage undershoot amplitude reached 28.8 mV, because the excessive gas flow rate in the channel carried more water vapor out of the cell, causing the membrane to be relatively dry due to insufficient water activity in the MEA. As the operating time increased and the water content in the membrane increased, the cell voltage gradually increased. It is noteworthy that there is a sudden change in voltage just before the voltage drop in fuel cells, which could be attributed to the redistribution of reactant gases and water within the MEA. However, overall, the relationship between the voltage fluctuation amplitudes in fuel cells after dynamic loading can be summarized as follows: 600 rpm > 400 rpm > 200 rpm. As shown in Fig. 7 (b), when the cell operates steadily for 5 min before load change, the cell performance was slightly reduced, indicating the existence of localized liquid water blockage inside the fuel cell. The voltage undershoot amplitude during load variation was much larger than in the previous case, reaching 97 mV, but the performance of the fuel cell quickly recovered. The results indicate that the hydration state of the membrane inside the fuel cell has a significant impact on the dynamic response during load variation for the dead-ended operation of hydrogen–oxygen fuel cells.

3.2.2. *Effect of recirculation pump speed of anode side*

The recirculation system is a process that transports unreacted reactants back to the input port, thereby minimizing the waste of reactants. Compared with the dead-end mode, the recirculation system does not require periodic purging operations, which enables more stable and durable operation. Depending on the operational requirements of the fuel cell, the recirculation system can be installed at the cathode or anode.

Fig. 8 depicts the voltage response curve from 200 mA·cm^{-2} to 1600 mA·cm^{-2} under different pump speeds of the anode side. At this time, the purging valve of the cathode outlet is always in a closed state. It can be seen from the results that the dynamic response of the fuel cell

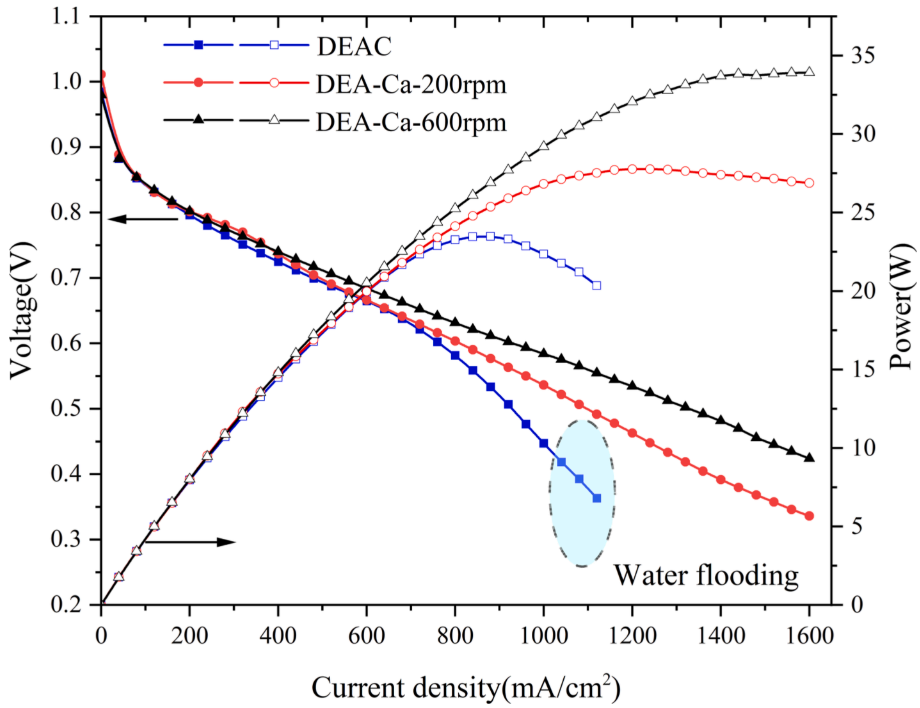

Fig. 5. Comparison of cell polarization curves at different recirculation pump speeds.

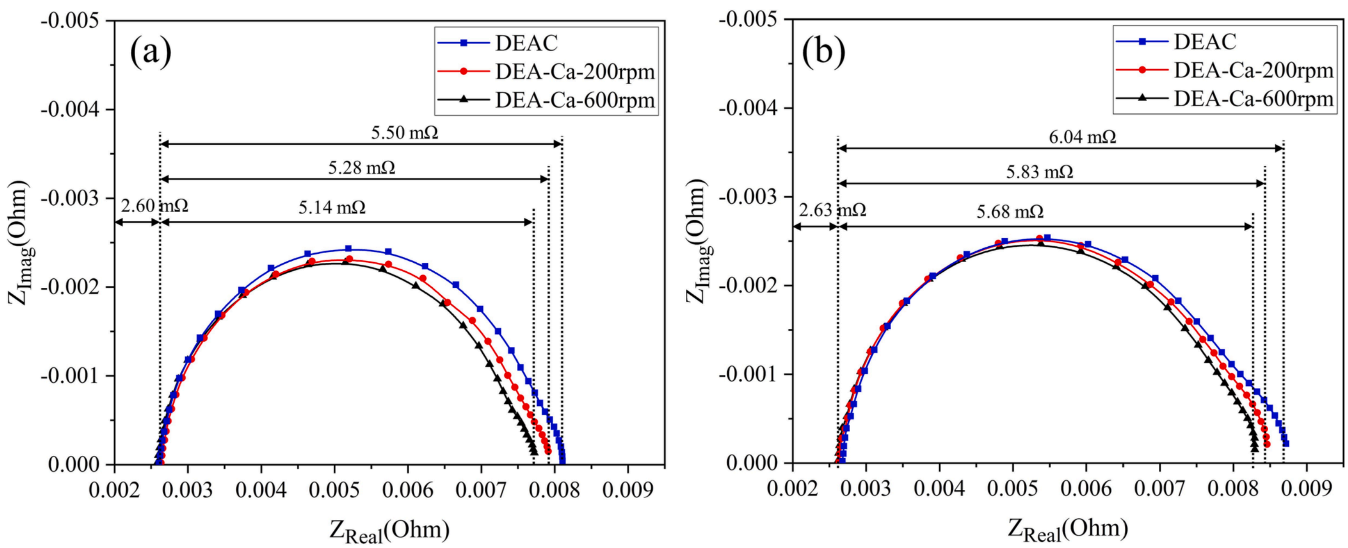

Fig. 6. EIS at different recirculation pump speeds. (a) 1000 mA·cm^{-2}, (b) 1600 mA·cm^{-2}.

voltage is more sensitive to the hydrogen flow rate in the anode channel. In other words, the higher the recirculation pump speed, the greater the voltage drop. When the anode recirculation pump speed reached 200, 400, and 600 rpm, the voltage undershoot amplitudes after load changes are 23.5 mV, 102.2 mV, and 166.4 mV, respectively. This is different from the phenomenon when the recirculation pump is assembled on the cathode side. During the operation of the fuel cell, compared with the rich water condition of the cathode, the water vapor content in the anode is relatively small. When the pump speed increases, the water vapor content brought out by convective mass transfer increases accordingly, leading to the deterioration of the hydration state of the membrane on the anode side. The water content in the membrane cannot be replenished in time under variable loading conditions, resulting in a large voltage undershoot amplitude. However, as the operating time increases, the performance of the fuel cell gradually recovers.

3.3. Mass transfer enhancement under current loading

During the operation of a PEMFC, activation polarization, ohmic polarization, and concentration polarization are the main causes of voltage undershoot [37]. Activation polarization and Ohmic polarization are related to the physical properties and initial conditions of the fuel cell, while concentration polarization is related to the structure and reaction process of the fuel cell. However, concentration polarization can be improved by enhancing the mass transport of reactant gases and water [38]. In addition, concentration polarization is also a major factor causing voltage undershoot during load changes in fuel cells. Currently,

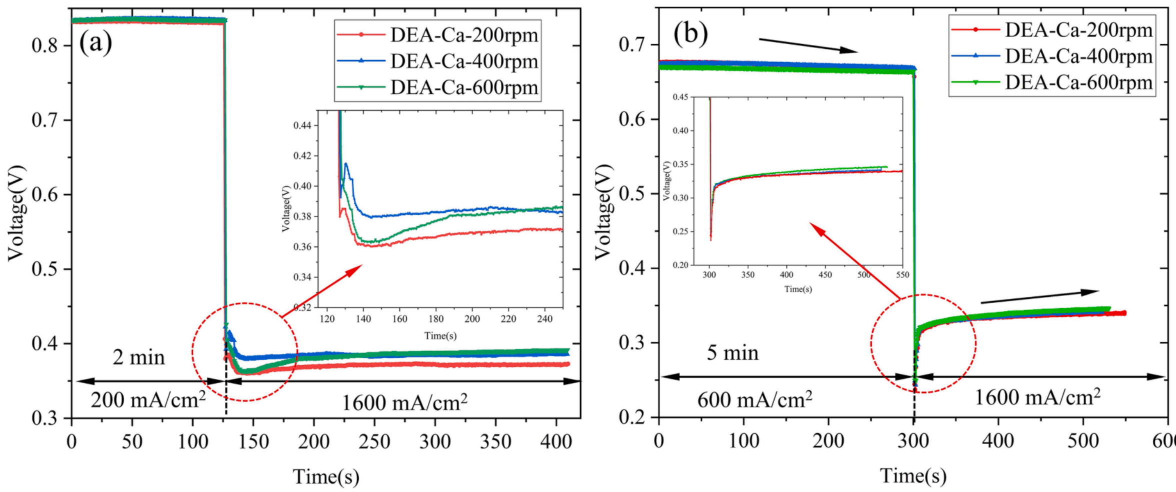

Fig. 7. Voltage response curve from 200 mA·cm^{-2} to 1600 mA·cm^{-2} under different pump speeds of cathode side. (a) 2 min continuous operation before loading, (b) 5 min continuous operation before loading.

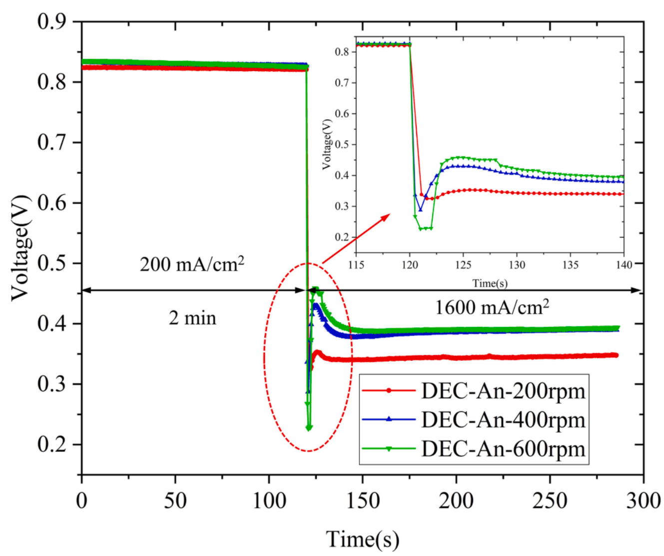

Fig. 8. Voltage response curve from 200 mA·cm^{-2} to 1600 mA·cm^{-2} under different pump speeds of anode side.

research has found that increasing the flow rate of reactant gases and adding turbulence promoters in the flow channels are effective ways to enhance mass transfer in fuel cells [39–41].

Therefore, we designed a flow channel structure with semi-circular small obstacles and tested the polarization curve and dynamic performance of the fuel cell assembly with this obstacle channel. The comparison of cell polarization curves at different recirculation pump speeds of the cathode side equipped with an obstacle channel is shown in Fig. 9. At low current densities, the output power of the cells with different flow channel structures changes slightly. As the current density increases, the single cell output power with obstacle channel was significantly improved, especially when the current density is higher than 1000 mA·cm^{-2}. For a conventional straight channel, the output power of the cell first increases and then decreases with the increase of current density, reaching the maximum at a current density of 1300 mA·cm^{-2}. As for the fuel cell with an obstacle channel, the inflection point of the change in cell power begins to move towards larger current densities. It can be seen that the turbulent structure in the fuel cell flow channel can greatly improve the mass transfer and diffusion of fuel gas, and alleviate the concentration loss under high current density. In addition, increasing the recirculation pump speed will increase the gas flow rate in the channel, and consequently improve the cell performance. The EIS maps at different recirculation pump speeds of the cathode side equipped with an obstacle channel are shown in Fig. 10. The trend of EIS curves at different current densities is almost the same, except that the polarization impedance value slightly increases at a current density of 1600 mA·cm^{-2}.

The voltage response curve from 200 mA·cm^{-2} to 1600 mA·cm^{-2} under different pump speeds of the cathode side equipped with an obstacle channel is shown in Fig. 11. Firstly when the fuel cell was abruptly loaded to a current density of 1600 mA·cm^{-2}, the voltage will decrease due to gas starvation in a short period of time, and this phenomenon will be alleviated with an increase in the speed of the recirculation pump. Subsequently, due to the effect of electric drag under high current density, the anode requires a large number of water molecules to satisfy the combination and transmission of protons. As the water desorption rate of the proton exchange membrane is much higher than the adsorption rate, when the loading current is too large and the supply rate of the anode water cannot meet its consumption rate, anode dehydration will occur [33]. After completion of the loading, the fuel cell will operate at a high current density and reach a dynamic equilibrium of water inside. However, when the speed of the recirculation pump is too high, the water inside the cell will be taken out in large quantities, which will be unfavorable to the wetting of the membrane electrode assembly. It can be seen from the results that when the speed of the recirculation pump is maintained at 400 rpm, the initial gas starvation and membrane hydration can be greatly alleviated.

3.4. Performance degradation under continuous cyclic loading

The voltage response characteristics of hydrogen–oxygen PEMFC under different configurations have been studied in the previous section. To further investigate the tolerance of voltage degradation to different fuel cell configurations, the performance and catalyst degradation rate of cells under different gas management strategies were analyzed (Three cases: DEAC mode, DEC with anode recirculation of 400 rpm, DEA with cathode recirculation of 400 rpm). The cyclic loading in this study was achieved by changing the current density from 200 mA·cm^{-2} to 1600 mA·cm^{-2}. The loading time between the two current densities was 1 s, and the loading amplitude was 1400 mA·cm^{-2}. The performance change of PEMFC was analyzed by recording the output voltage. Each current density was maintained for 30 s. After 500 dynamic loading cycles, the performance of PEMFC was studied and compared by obtaining the polarization curve, and EIS map.

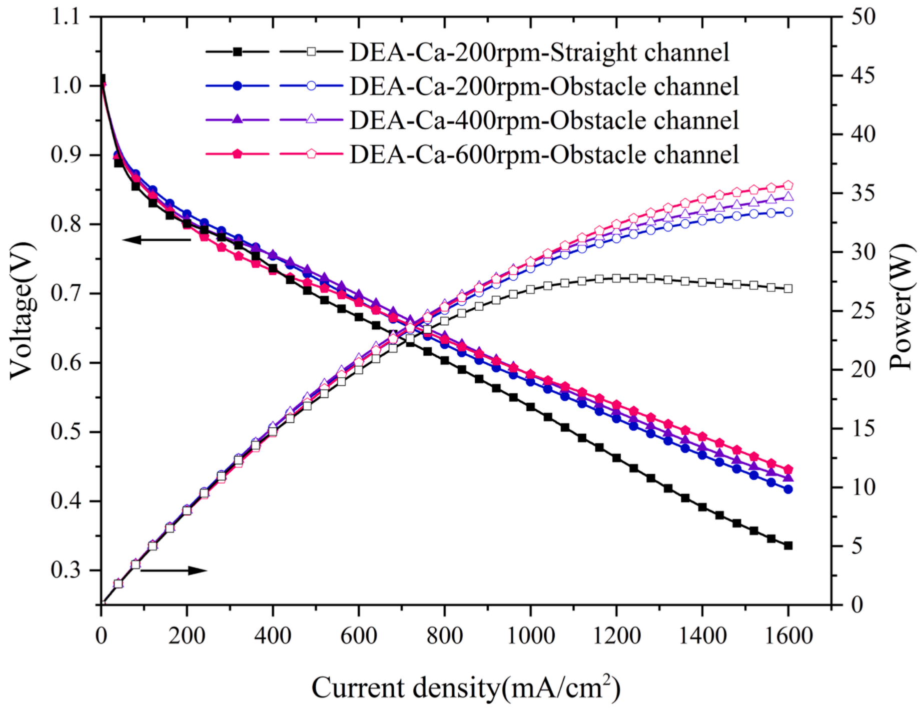

Fig. 9. Comparison of cell polarization curves at different recirculation pump speeds of cathode side equipped with obstacle channel.

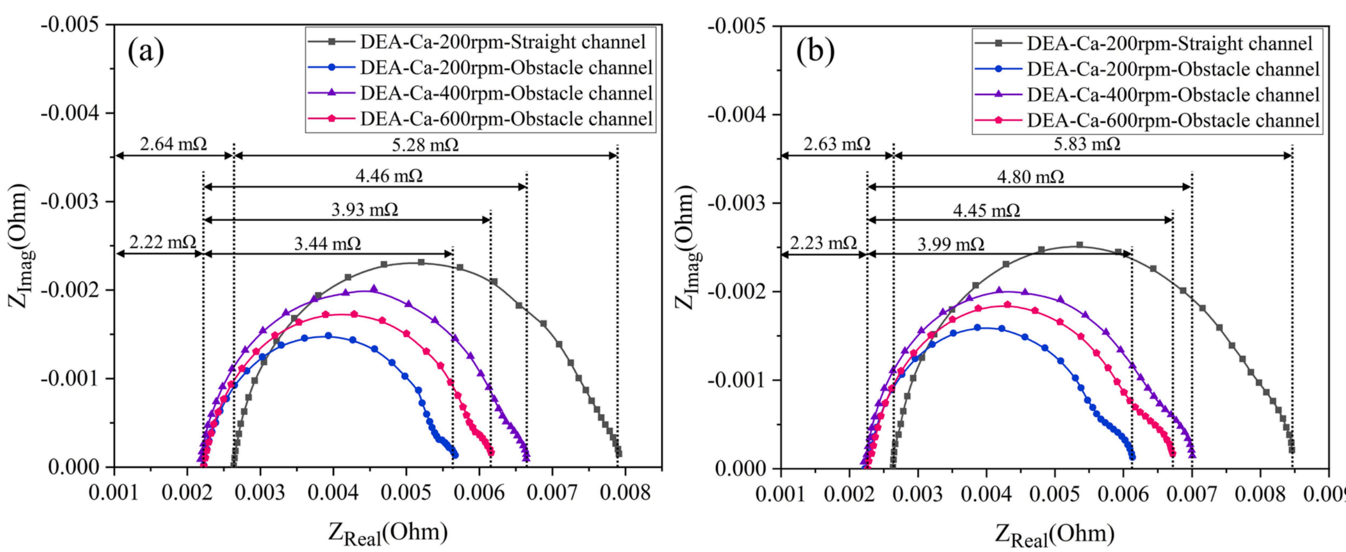

Fig. 10. EIS at different recirculation pump speeds of cathode side equipped with obstacle channel. (a) 1000 mA·cm^{-2}, (b) 1600 mA·cm^{-2}.

The voltage evolution under different fuel cell configurations is shown in Fig. 12. During dynamic loading, the voltage will rapidly decrease and undershoot. The voltage undershoot is the largest when the anode is configured with a recirculation pump, reaching 0.17 V. This is attributed to the fact that fuel cells are more sensitive to hydrogen starvation [28]. Anode exhaust gas recirculation will accelerate the flow rate of reaction gas, but will play a negative role in the diffusion of hydrogen to GDL direction. In addition, under the DEAC mode, the steady-state voltage will decrease with time after each cyclic loading, until it reaches the gas purging period. A certain amount of liquid water will accumulate in the cathode and anode of the fuel cell in DEAC mode, which will hinder gas transmission, and the voltage will gradually decline after several cycles of loading. Therefore, the internal water management situation of the cell is an important influencing factor for the dynamic response characteristics of hydrogen–oxygen PEMFCs. The addition of cathode pump will greatly alleviate the cathode flooding, so that the gas starvation in the loading process is mitigated, and the amplitude of voltage drop is reduced to only about 0.05 V, as shown in Fig. 12 (c).

After continuous the cyclic loading experiment, the polarization curve of PEMFC was recorded to characterize the performance change of the cell. The MEAs test before and after the experiments were carried out by 850e test platform under the test conditions of Stoi An/Ca = 2/3, operating temperature 60℃ and back pressure 80 kPa. As shown in Fig. 13, the output power and voltage of PEMFC significantly decrease under the DEAC operating strategy. When the current density is 1600 mA·cm^{-2}, the voltage decreases from 0.444 V to 0.340 V, and the performance drops by 23.42%. However, when the exhaust gas recirculation device is installed on the cathode side of the fuel cell, the voltage only decreases from 0.444 V to 0.428 V at a current density of 1600

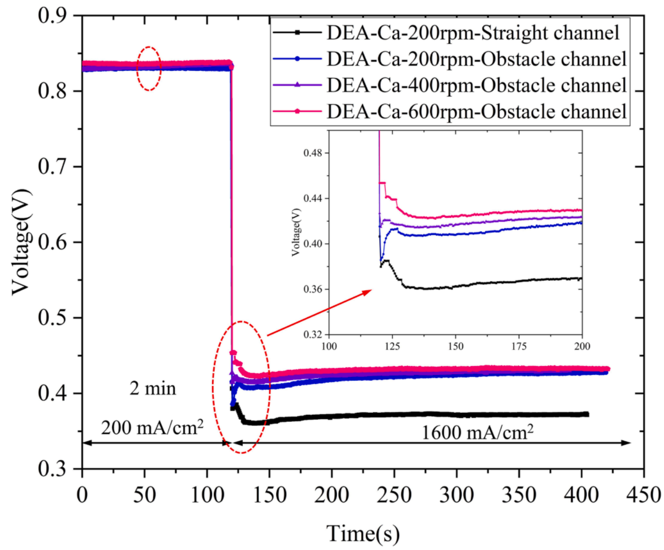

Fig. 11. Voltage response curve from 200 mA·cm^{-2} to 1600 mA·cm^{-2} under different pump speeds of cathode side equipped with obstacle channel.

mA·cm^{-2}, with a performance decrease of only 3.6%. In addition, compared with the recirculation pump equipped on the cathode side, the recirculation pump equipped on the anode side causes more severe performance degradation of the PEMFC.

To further investigate the effect of gas starvation during continuous cyclic loading of the PEMFC, Fig. 14 shows the EIS curves after the experiment. The Nyquist plot shows that after the cyclic loading experiment, the ohmic resistance, which represents the impedance of ion and electron transport inside the cell and the contact impedance between various components of the PEMFC, hardly changed. However, the polarization resistance at low frequencies varied significantly under different experimental conditions. Under a current density of 1000 mA·cm^{-2}, the impedance values corresponding to the three experimental conditions increased from 2.57 mΩ to 4.86 mΩ, 5.64 mΩ, and 5.88 mΩ, respectively. The increase in polarization resistance represents a decline in the mass transfer capacity of the membrane electrode assembly, which is the main cause of cell performance degradation.

The CV curve is an important method to characterize the electrochemical surface area (ECSA) of PEMFC catalysts. Fig. 15 shows the CV curves before and after the cyclic loading experiment. The ECSA serves as a quantitative measure of the effective catalytic area within the CL. It can be calculated using the following formula:

$$ECSA = \frac{Q_H}{[Pt] \times 0.21} \tag{1}$$

where Q_H is the charge associated with the hydrogen adsorption and desorption peaks in the CV curve, $[Pt]$ is the mass of platinum per unit area. The findings from the study reveal significant insights regarding the impact of different operating strategies on the performance of the fuel cell. To quantify the changes in ECSA during the experiment, the charge area of the hydrogen evolution peak in the CV was measured. The calculation results show that ECSA decreases from 92.36 m^2·g^{-1} to 74.62 m^2·g^{-1} during the DEAC operating strategy, with a decrease rate of 19.21%. However, in the anode recirculation experiment, ECSA decreased from 92.25 m^2·g^{-1} to 85.32 m^2·g^{-1}, with a decrease rate of 7.51%. In the cathode recirculation experiment, ECSA decreased from 92.30 m^2·g^{-1} to 90.79 m^2·g^{-1}, with a decrease rate of only 1.64% in ECSA. The CV results show that the fuel cell operating in the DEAC operating strategy is more likely to cause MEA degradation, while exhaust gas recirculation can significantly alleviate gas starvation and water transfer during the cyclic loading process of the fuel cell, and the cathode recirculation is the primary mitigation strategy to be considered.

4. Conclusions

In this study, the dynamic characteristics of the H_2/O_2 PEMFC with a recirculation subsystem were studied. And the impact of flow channel configuration under the exhaust gas recirculation mode on cell performance and dynamic loading characteristics was also investigated. The conclusions are summarized as follows:

1) The primary issue to be addressed during the loading process of H_2/O_2 PEMFCs operated in the DEAC mode is the blockage of the GDL gas transport channel caused by accumulated liquid water. The cathode recirculation pumps of 600 rpm can improve the cell performance by 30.6% compared to the DEAC mode at a current density of 1000 mA·cm^{-2}. However, excessive pump speed can lead to an increase in voltage undershoot during the loading process.
2) For the fuel cell with an obstacle channel, the inflection point of the change in cell power tends to move towards larger current densities. The turbulent structure in the fuel cell flow channel can greatly improve the mass transfer and diffusion of fuel gas during the loading process, and alleviate the concentration loss under high current density.
3) After the continuous cyclic loading experiment, the cell performance with DEAC mode decreased by 23.42% at 1600 mA·cm^{-2}, and the ECSA decreased by 19.21%. However, the cell performance with cathode recirculation decreased by 3.6% at 1600 mA·cm^{-2}, and the ECSA decreased by 1.64%. The fuel cell operating in the DEAC operating strategy is more likely to cause MEA degradation, while exhaust gas recirculation can significantly alleviate gas starvation and water transfer during the cyclic loading process of the fuel cell, and the cathode recirculation is the primary mitigation strategy to be considered.

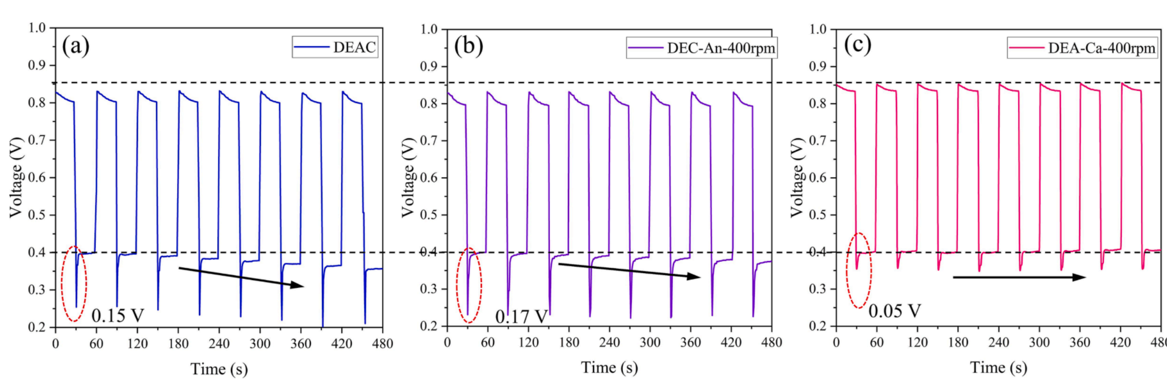

Fig. 12. Voltage evolution during continuous cyclic loading. (a) DEAC mode, (b) DEC-An-400 rpm, (c) DEA-Ca-400 rpm.

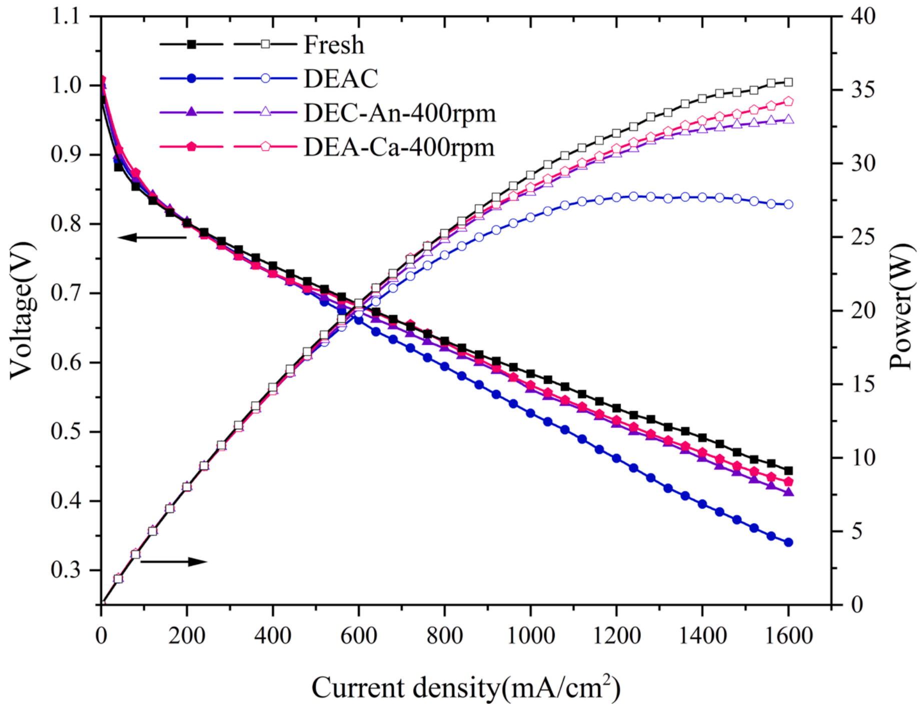

Fig. 13. Performance degradation of the hydrogen–oxygen PEMFC after continuous cyclic loading.

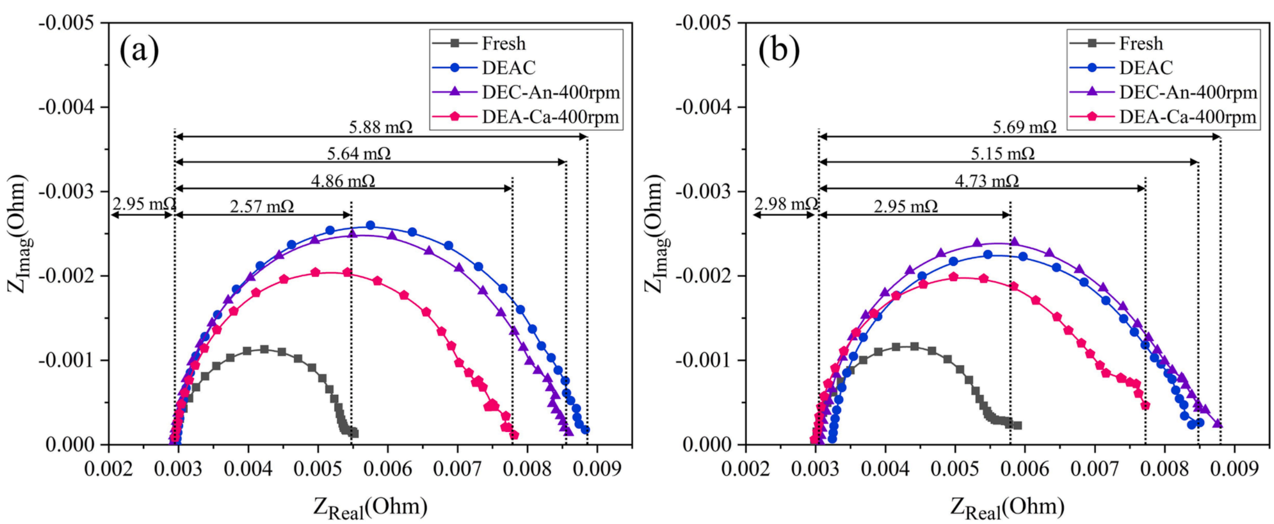

Fig. 14. Comparison of EIS curves after continuous cyclic loading. (a) 1000 mA·cm^{-2}, (b) 1600 mA·cm^{-2}.

The present study primarily focuses on the investigation of the dynamic characteristics of hydrogen–oxygen fuel cells at the single-cell level. However, the dynamic behavior of the entire fuel cell stack, and even the whole system, is more complex yet holds greater engineering value. Therefore, further research is necessary to extend the relevant experimental exploration to the entire stack system, including issues related to the consistency between individual cells during the dynamic loading process of high-power stacks and the performance degradation mechanism of the stack. In addition, during the operation of fuel cells, adverse conditions such as dehydration and flooding can occur due to inappropriate operating conditions and loading strategies. Timely identification and resolution of these issues are of significant importance in maintaining their excellent load-variable characteristics. Therefore, the next step should focus on the research of online diagnostic techniques for fuel cell and their corresponding countermeasures to ensure the efficient and stable operation of fuel cells.

CRediT authorship contribution statement

Yang Liu: . **Junjie Zhao:** Conceptualization. **Zhengkai Tu:** . **Siew Hwa Chan:** .

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

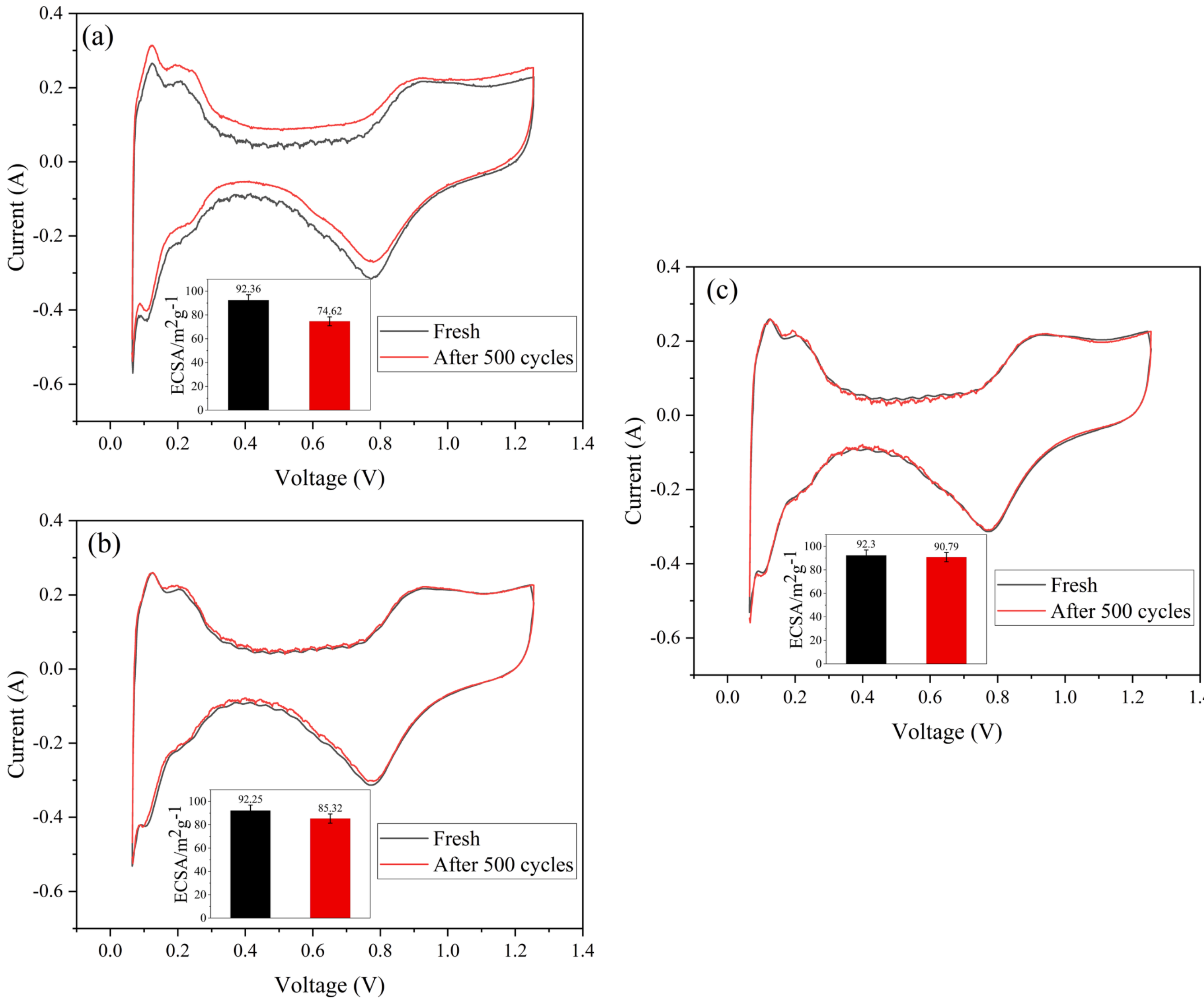

Fig. 15. CV curves after continuous cyclic loading experiment. (a) DEAC mode, (b) DEC-An-400 rpm, (c) DEA-Ca-400 rpm.

Data availability

No data was used for the research described in the article.

Acknowledgments

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References

[1] Zhang G, Qu Z, Tao WQ, et al. Porous Flow Field for Next-Generation Proton Exchange Membrane Fuel Cells: Materials, Characterization, Design, and Challenges. Chem Rev 2023;123(3):989–1039.
[2] Wilberforce T, Amiri A. Comparative study on bipolar plate geometry designs on the performance of proton exchange membrane fuel cells. Fuel 2023;346:128389.
[3] Mitra U, Arya A, Gupta S. A comprehensive and comparative review on parameter estimation methods for modelling proton exchange membrane fuel cell. Fuel 2023; 335:127080.
[4] Duan Y, Liu H, Zhang W, et al. Materials, components, assembly and performance of flexible polymer electrolyte membrane fuel cell: A review. J Power Sources 2023;555:232369.
[5] Liu Y, Tu Z, Chan SH. Water management and performance enhancement in a proton exchange membrane fuel cell system using optimized gas recirculation devices. Energy 2023;279:128029.
[6] Liu Y, Tu Z, Chan SH. Performance Evaluation and Degradation Mechanism for Proton Exchange Membrane Fuel Cell with Dual Exhaust Gas Recirculation. Adv Energy Sustainability Res 2023;4(7):2200180.
[7] Liu Y, Tu Z, Chan SH. Performance analysis and dynamic characteristics of a proton exchange membrane fuel cell with dual recirculation pumps for air-free applications. J Power Sources 2023;566:232926.
[8] Shen J, Du C, Yan F, et al. Experimental study on the dynamic performance of a power system with dual air-cooled PEMFC stacks. Appl Energy 2022;326:120025.
[9] Lin R, Ren YS, Lin XW, et al. Investigation of the internal behavior in segmented PEMFCs of different flow fields during cold start process. Energy 2017;123:367–77.
[10] Yu Y, Yuan X-Z, Li H, et al. Current mapping of a proton exchange membrane fuel cell with a segmented current collector during the gas starvation and shutdown processes. Int J Hydrogen Energy 2012;37(20):15288–300.
[11] Zhao J, Jian QF, Huang ZP. Visualization study on enhancing water transport of proton exchange membrane fuel cells with a dead-ended anode by generating fluctuating flow at anode compartment. Energ Conver Manage 2020;206:112477.
[12] Weiland M, Boillat P, Oberholzer P, et al. High resolution neutron imaging for pulsed and constant load operation of passive self-breathing polymer electrolyte fuel cells. Electrochim Acta 2013;87:567–74.
[13] Meng K, Chen B, Zhou H, et al. Experimentally investigation on current density distribution characteristics of hydrogen-oxygen proton exchange membrane fuel cells under dynamic loading. J Clean Prod 2023;393:136315.
[14] Jia F, Tian X, Liu F, et al. Oxidant starvation under various operating conditions on local and transient performance of proton exchange membrane fuel cells. Appl Energy 2023;331:120412.
[15] Zhang G, Shen S, Guo L, et al. Dynamic characteristics of local current densities and temperatures in proton exchange membrane fuel cells during reactant starvations. Int J Hydrogen Energy 2012;37(2):1884–92.

[16] Wang G, Huang F, Yu Y, et al. Degradation behavior of a proton exchange membrane fuel cell stack under dynamic cycles between idling and rated condition. Int J Hydrogen Energy 2018;43(9):4471–81.
[17] Chu T, Xie M, Yu Y, et al. Experimental study of the influence of dynamic load cycle and operating parameters on the durability of PEMFC. Energy 2022;239:122356.
[18] Chu T, Wang Q, Xie M, et al. Investigation of the reversible performance degradation mechanism of the PEMFC stack during long-term durability test. Energy 2022;258:124747.
[19] Zhong D, Lin R, Liu D, et al. Structure optimization of anode parallel flow field for local starvation of proton exchange membrane fuel cell. J Power Sources 2018;403:1–10.
[20] Jia F, Guo L, Liu H. Mitigation strategies for hydrogen starvation under dynamic loading in proton exchange membrane fuel cells. Energ Conver Manage 2017;139:175–81.
[21] Bai X, Jian Q, Huang B, et al. Hydrogen starvation mitigation strategies during the start-up of proton exchange membrane fuel cell stack. J Power Sources 2022;520:230809.
[22] Wang Y, Li H, Feng H, et al. Simulation study on the PEMFC oxygen starvation based on the coupling algorithm of model predictive control and PID. Energ Conver Manage 2021;249.
[23] Ding R, Zhang S, Chen Y, et al. Application of Machine Learning in Optimizing Proton Exchange Membrane Fuel Cells: A Review. Energy and AI 2022;9:100170.
[24] Wang J, Jiang H, Chen G, et al. Integration of multi-physics and machine learning-based surrogate modelling approaches for multi-objective optimization of deformed GDL of PEM fuel cells. Energy and AI 2023;14:100261.
[25] Wang J, Ding R, Cao F, et al. Comparison of state-of-the-art machine learning algorithms and data-driven optimization methods for mitigating nitrogen crossover in PEM fuel cells. Chem Eng J 2022;442:136064.
[26] Wang B, Yang Z, Ji M, et al. Long short-term memory deep learning model for predicting the dynamic performance of automotive PEMFC system. Energy and AI 2023;14:100278.
[27] Rui Z, Ding R, Hua K, et al. Design of proton exchange membranes with high durability for fuel cells: From the perspective of machine learning. J Membr Sci 2023;683:121831.
[28] Meng K, Chen B, Zhou H, et al. Experimental investigation on voltage response characteristics of hydrogen-oxygen proton exchange membrane fuel cells under gas starvation. Energ Conver Manage 2022;268:115973.
[29] Meng K, Chen B, Zhou H, et al. Investigation on degradation mechanism of hydrogen–oxygen proton exchange membrane fuel cell under current cyclic loading. Energy 2022;242:123045.
[30] Chen W, Chen B, Meng K, et al. Experimental study on dynamic response characteristics and performance degradation mechanism of hydrogen-oxygen PEMFC during loading. Int J Hydrogen Energy 2023;48(12):4800–11.
[31] Fan L, Xing L, Tu Z, et al. A breakthrough hydrogen and oxygen utilization in a H2–O2 PEMFC stack with dead-ended anode and cathode. Energ Conver Manage 2021;243:114404.
[32] Alizadeh E, Khorshidian M, Saadat SHM, et al. The experimental analysis of a dead-end H2/O2 PEM fuel cell stack with cascade type design. Int J Hydrogen Energy 2017;42(16):11662–72.
[33] Huang Z, Shen J, Chan SH, et al. Transient response of performance in a proton exchange membrane fuel cell under dynamic loading. Energ Conver Manage 2020;226:113492.
[34] Celik E, Karagoz I. Investigation of the effects of intermediate reservoirs and intermediate feedings applications on the performance of proton exchange membrane fuel cells. Fuel 2023;339:126975.
[35] Zhao L, Yuan H, Xie J, et al. Inconsistency evaluation of vehicle-oriented fuel cell stacks based on electrochemical impedance under dynamic operating conditions. Energy 2023;265:126162.
[36] Xia ZF, Chen HC, Shan WC, et al. Behavior of current distribution evolution under reactant starvation conditions based on a single polymer electrolyte membrane fuel cell (PEMFC) with triple-serpentine flow field: An experimental study. Int J Hydrogen Energy 2023;48(36):13650–68.
[37] Xu L, Fang C, Li J, et al. Nonlinear dynamic mechanism modeling of a polymer electrolyte membrane fuel cell with dead-ended anode considering mass transport and actuator properties. Appl Energy 2018;230:106–21.
[38] Moein-Jahromi M, Kermani MJ. Three-dimensional multiphase simulation and multi-objective optimization of PEM fuel cells degradation under automotive cyclic loads. Energ Conver Manage 2021;231:113837.
[39] Shen J, Tu Z. Flow channel design in a proton exchange membrane fuel cell: From 2D to 3D. Int J Hydrogen Energy 2022;47(5):3087–98.
[40] Shen J, Tu ZK, Chan SH. Performance enhancement in a proton exchange membrane fuel cell with a novel 3D flow field. Appl Therm Eng 2020;164:114464.
[41] Shen J, Tu Z, Chan SH. Evaluation criterion of different flow field patterns in a proton exchange membrane fuel cell. Energ Conver Manage 2020;213:112841.